Tutorial - 2

identify the ~~f~~ surface of each of the following eqn :- (a) $r = 5$

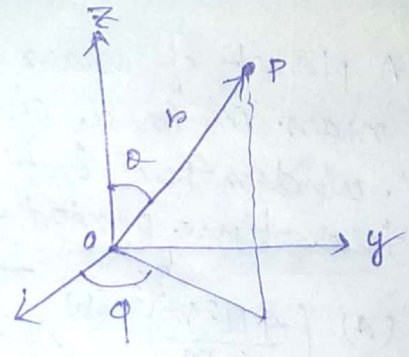
(b) $\theta = \pi/3$.

(a) $r = 5$ → just convert it into the cartesian C.S.

$r = 5$

$\Rightarrow r^2 = 25$

$\Rightarrow \boxed{x^2 + y^2 + z^2 = 5^2}$ → eqn of sphere of radius 5 units, centered at origin.



NB: Try to visualise.

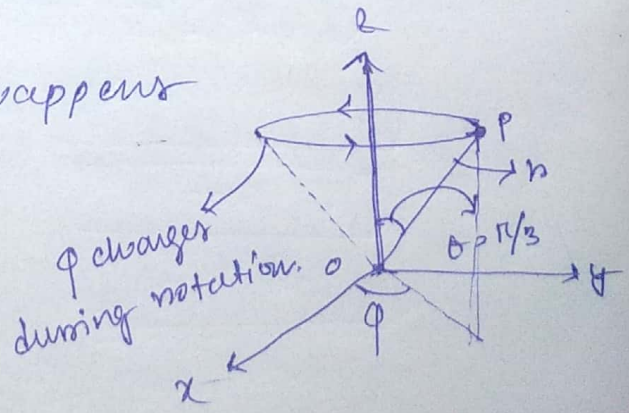
Here, r is fixed, but ~~$\theta = \pi/$~~ θ & φ is not fixed. So, we can rotate a point (P) as much away from z axis and around the z axis but the point should be always at a fixed distance $r = 5$ from origin,

which indicates, it is nothing but a sphere of radius 5 units centered at origin.

(b) $\theta = \pi/3$

This indicates, no matter how far we go from origin (r) and how much we rotate a point about the z axis, the point always makes an angle $\theta = \pi/3$ with the z axis.

This is what exactly happens in a cone.



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(2) Convert the point $(\sqrt{6}, \pi/4, \sqrt{2})$ from cylindrical to spherical co-ordinates.

→ in cylindrical C.S → $(\rho, \phi, z) \equiv (\sqrt{6}, \pi/4, \sqrt{2})$

with, $\rho = \sqrt{6}$, $\phi = \pi/4$, $z = \sqrt{2}$

cylindrical C.S → cartesian C.S → spherical C.S

Relation betn cylindrical CS and cartesian C.S —

$$\left.\begin{aligned} x &= \rho\cos\phi \\ y &= \rho\sin\phi \\ z &= z. \end{aligned}\right\} \quad (1)$$

So, $x^2+y^2+z^2 = \rho^2+z^2$

$\Rightarrow r^2 = \rho^2+z^2$

And, rel^n betn cartesian C.S to spherical C.S —

$$\left.\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \end{aligned}\right\} \quad (2)$$

So, comparing (1) & (2) we can say that, from cylindrical to spherical,

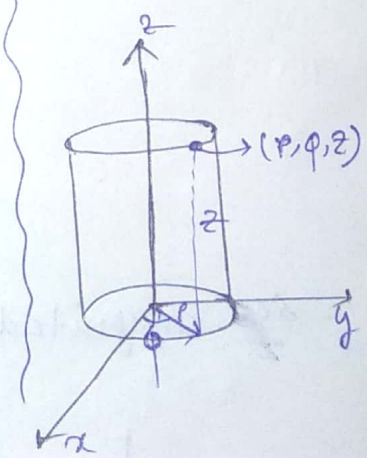
'ϕ' - remains same.

So, we have to find 'r' & 'θ'.

Now (1) → $x^2+y^2+z^2 = \rho^2+z^2$

$\Rightarrow r^2 = (6)+(2) = 8$

$\Rightarrow \boxed{r = \sqrt{8} = 2\sqrt{2}}$



Now $z = r\cos\theta$

$\Rightarrow \sqrt{2} = 2\sqrt{2}\cos\theta$

$\Rightarrow \cos\theta = 1/2 \Rightarrow \boxed{\theta = \pi/3}$

So, the transformed point is $\equiv (2\sqrt{2}, \pi/3, \pi/4)$

(Ans)

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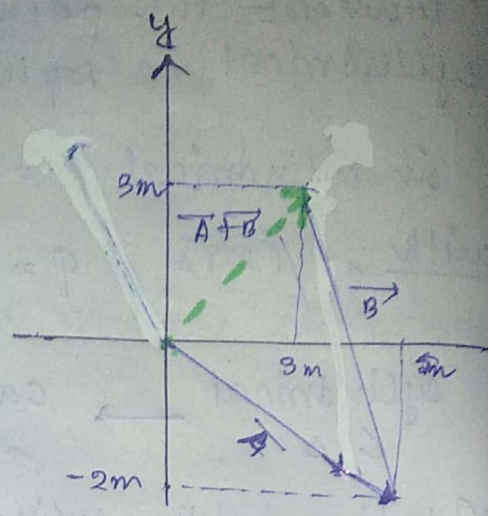
3. A group of vectors are shown in the figure below →

(a) Find direction & magnitude of $\vec{A}$.

(b) Find direction & mag. of $\vec{B}$.

(c) Find the dirn and mag. of $(\vec{A}+\vec{B})$

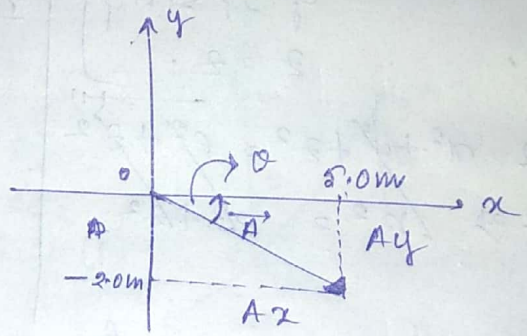


⇒ (a) $A_x = (\text{final} - \text{initial})$

$= (5-0)\,m$

$= 5m$

$A_y = (-2-0)\,m = -2m$



$\tan\theta = \frac{A_y}{A_x}$ (measured in anti-clock-wise dirn w.r.to x axis)

$\Rightarrow \tan\theta = \frac{-2m}{5m} = -2/5$

$\Rightarrow \theta = \tan^{-1}(-2/5) = -21.80^\circ$

$\cancel{\Rightarrow \theta = \pi - \tan^{-1}(2/5)}$

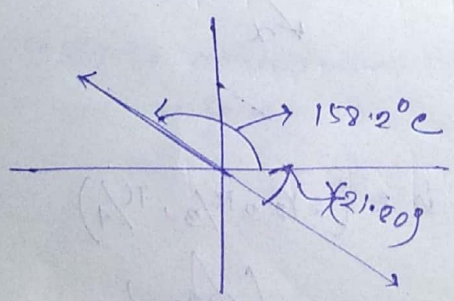
$\cancel{\Rightarrow \theta = \pi - 21.80}$

$\Rightarrow \boxed{\theta = -21.80^\circ}$

So, magnitude $|\vec{A}| = \sqrt{A_x^2 + A_y^2}$

$= \sqrt{25 + 4}\,m = \sqrt{29}\,m$

$\Rightarrow \boxed{|A| = 5.38\,m}$



$\boxed{\theta = -21.80^\circ}$

$\cancel{\theta = 158.2}$

or, $\boxed{\theta = 158.2^\circ}$

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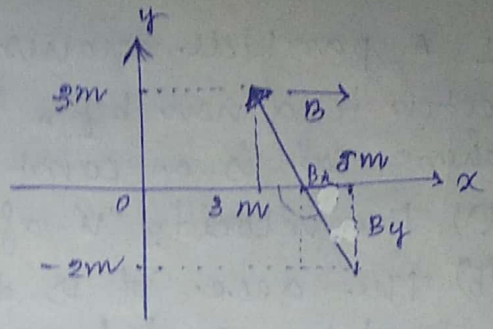
(b) $B_x = (3-2)m = -2m$

$B_y = (3-(-2))m = 5m$



$\therefore |\vec{B}| = \sqrt{B_x^2 + B_y^2}$

$= \sqrt{4+25}\ m$

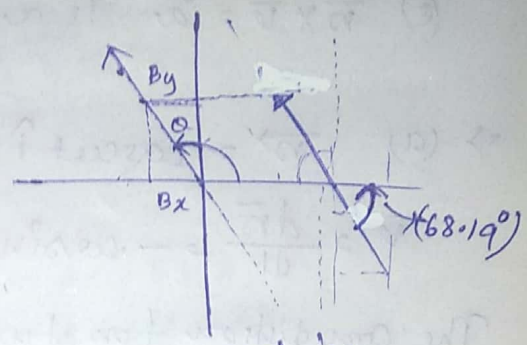
$= \sqrt{29}\ m = 5.38m$

$\tan\theta = \frac{B_y}{B_x} = \left(\frac{5}{-2}\right)$

$\Rightarrow \theta = \tan^{-1}(-5/2) =$ ~~[illegible]~~

$= -68.19^\circ = 111.81^\circ$

$\Rightarrow \boxed{\theta = 111.81^\circ}$



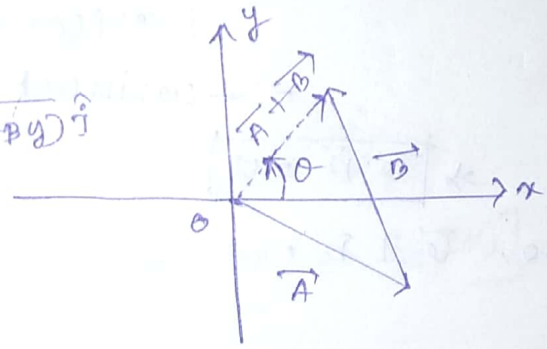
↳ w.r.t. x axis, measured in anti-clockwise dir

(c)

$\vec{A} + \vec{B} = \overline{(A_x + B_x)}\hat{i} + \overline{(A_y + B_y)}\hat{j}$

$= (2-2)\hat{i} + (-2+5)\hat{j}$

$= 3\hat{i} + 3\hat{j}$



$|\vec{A} + \vec{B}| = \sqrt{9+9} = 3\sqrt{2}\ m$

$\tan\theta = \frac{|(\vec{A}+\vec{B})|_x}{|\vec{A}+\vec{B}|_y} = \frac{3}{3} = 1$

$\Rightarrow \boxed{\theta = 45^\circ}$

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4. A particle moves so that its position vector is given by, $\vec{r} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$, where, 'ω' is a const. show that,

(a) the velocity 'v' of the particle is ⊥ to $\vec{r}$,

(b) the accel. 'a' is directed towards the origin and has the magnitude, proportional to the distance from origin.

(c) $\vec{r} \times \vec{v} = \vec{a}$ is a const. vector.

→ (a) $\vec{r} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$

$\vec{v} = \frac{d\vec{r}}{dt} = -\omega\sin\omega t\,\hat{i} + \omega\cos\omega t\,\hat{j}$

The condition for two vectors to be ⊥ is -

$\boxed{\vec{a}\cdot\vec{b} = 0}$

Here for $\vec{v}\cdot\vec{r} = (-\omega\sin\omega t\,\hat{i} + \omega\cos\omega t\,\hat{j})$
$\cdot(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$

$= -\omega\sin\omega t\cos\omega t + \omega\sin\omega t\cdot\cos\omega t$

$\Rightarrow \boxed{\vec{v}\cdot\vec{r} = 0}$

So, $\vec{v} \perp \vec{r}$,

(b) $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

$= -\omega^2\cos\omega t\,\hat{i} - \omega^2\sin\omega t\,\hat{j}$

$= -\omega^2(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$

$= -\omega^2\vec{r}$

$|\vec{a}| = \omega^2|\vec{r}|$

So, '−' sign indicated the vector is directed towards origin.

and, $|\vec{a}| \propto |\vec{r}|$

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(c) $\bar{r} \times \bar{v}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega\sin\omega t & \omega\cos\omega t & 0 \end{vmatrix}$$

$$= \hat{i}(0-0) + \hat{j}(0-0) + \cancel{\hat{k}(\omega\sin\omega t}$$

$$+ \hat{k}(\omega\cos^2\omega t + \omega\sin^2\omega t)$$

$$= \omega\hat{k} = \text{const.}$$

(5) Prove that, $\bar{A}\cdot(\bar{B}\times\bar{C}) = \bar{B}\cdot(\bar{C}\times\bar{A})$
$= \bar{C}\cdot(\bar{A}\times\bar{B})$

$$\Rightarrow \bar{A}\cdot(\bar{B}\times\bar{C}) = \bar{A}\cdot\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$= (A_x\hat{i} + A_y\hat{j} + A_z\hat{k})\cdot[\hat{i}(B_yC_z - B_zC_y)$$

$$+ \hat{j}(B_zC_x - B_xC_z) + \hat{k}(B_xC_y - B_yC_x)]$$

$$= A_x(B_yC_z - B_zC_y) + A_y(B_zC_x - B_xC_z)$$

$$+ A_z(B_xC_y - B_yC_x)$$

$$= \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$ → Det. of a matrix

interchanging of two rows changes the sign.

Now

$$-\begin{vmatrix} B_x & B_y & B_z \\ A_x & A_y & A_z \\ C_x & C_y & C_z \end{vmatrix} = +\begin{vmatrix} B_x & B_y & B_z \\ C_x & C_y & C_z \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \bar{B}\cdot(\bar{C}\times\bar{A})$$

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$$\begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = -\begin{vmatrix} C_x & C_y & C_z \\ B_x & B_y & B_z \\ A_x & A_y & A_z \end{vmatrix}$$

$$= +\begin{vmatrix} C_x & C_y & C_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \vec{C} \cdot (\vec{A} \times \vec{B})$$

<u>So,</u> $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$

(Proved)

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